DS2 and Dhalion vs. uniform DoP 2 at the same input rate (L = 100k, M = 500k, H = 1M events/s), m510

		Flink						Storm						Spark						Kafka S.						RisingW.					
		DS2			Dhalion			DS2			Dhalion			DS2			Dhalion			DS2			Dhalion			DS2			Dhalion		
		L	M	H	L	M	H	L	M	H	L	M	H	L	M	H	L	M	H	L	M	H	L	M	H	L	M	H	L	M	H
CPU	Sentiment Analysis (SA)	12	9.1	13	1.1	0.97	0.98	n/a	n/a	n/a	n/a	n/a	n/a	0.97	1.7	1.6	0.98	2.3	2.4	1	1	n/a	1	1	n/a	1	1	1	1	0.98	1
CPU	Traffic Monitoring (TM)	8.3	44	67	8.4	41	64	0.65	0.46	0.081	0.55	0.76	1.3	0.72	1.3	3.7	5.7	7.9	14	337	n/a	n/a	280	320	779	1	1.1	1	1	0.76	1
CPU	Trending Topics (TT)	120	46	66	99	27	9.2	0.02	0.37	3.8	0.02	0.046	0.45	0.32	0.13	0.002	0.96	0.41	0.28	0.95	0.18	0.25	0.078	0.32	0.12	0.86	0.31	3.7	0.91	0.99	6.2
CPU	Machine Outlier (MO)	4	1.8	1.8	1.1	2.6	3.9	6.6	0.97	0.14	0.48	0.53	5	1	1	0.98	1	1.1	1.2	1	1.5	n/a	1	0.99	n/a	1.4	1.9	2.1	1.8	2.3	2.2
S+N State	Ad Analytics (AD)	5.2	2.2	2.2	5	1.9	1.8	31	3.2	n/a	0.4	0.9	n/a	0.53	0.15	0.12	1.1	0.9	0.48	34	n/a	n/a	57	54	67	0.97	1.4	1.2	1	1.5	1.2
S+N State	Click Analytics (CA)	3.9	1.4	1.4	3.7	1.2	1.2	1.2	0.84	n/a	0.29	1.1	n/a	0.98	0.22	0.49	0.98	0.68	0.56	363	n/a	n/a	399	393	326	1.1	n/a	1.1	1	n/a	1
S+N State	Linear Road (LR)	2.7	0.99	0.77	1.4	1.6	1.2	n/a	n/a	n/a	n/a	n/a	n/a	1	0.99	1.3	1	1	1.5	0.99	1	n/a	0.98	1	n/a	0.99	0.8	1.3	0.92	0.89	1.2
State+UDF	Log Analyzer (LA)	9.7	2.2	1.9	9	2.1	1.8	1.1	0.021	4.1	0.12	0.15	4.1	0.98	0.37	0.88	1	0.54	1.3	926	113	n/a	625	689	n/a	n/a	n/a	n/a	n/a	n/a	n/a
State+UDF	Bargain Index (BI)	2.6	1.1	0.79	3.2	2.8	0.79	1	0.33	n/a	0.58	4.5	n/a	1	0.51	0.79	1	0.89	0.8	71	60	186	83	89	107	0.86	1	1.3	1	0.67	0.77
State+UDF	Smart Grid (SG)	0.86	1.1	0.56	4.1	0.66	2	0.028	n/a	0.033	0.042	0.13	0.027	0.98	0.28	0.052	0.84	0.2	0.5	376	6292	n/a	317	0.077	n/a	1	0.36	0.24	1	0.35	0.23
State+UDF	Cloud Monitoring (GCM)	3.3	0.4	1.8	2.5	0.17	0.73	n/a	n/a	n/a	n/a	n/a	n/a	1	0.36	0.19	1	0.94	0.67	314	391	n/a	235	248	599	0.56	1.1	0.89	0.97	1.1	0.93
State+UDF	Spike Detection (SD)	0.002	0.53	n/a	0.097	0.5	n/a	n/a	n/a	n/a	n/a	n/a	n/a	0.98	0.15	0.52	0.95	0.69	0.8	371	40	24	399	99	89	1	1.3	1.1	0.98	0.98	1.1
Stateless	TPC-H (TPCH)	1	1	1.1	1	1	0.92	0.000	1.4	1.6	0.000	0.92	0.89	0.91	0.99	1.3	0.99	1	1.4	0.99	1	1.3	1	1	1.3	0.81	0.52	1.6	1	0.97	1.5
Stateless	Word Count (WC)	0.92	0.75	0.82	0.92	0.77	0.83	0.84	0.94	n/a	0.88	0.87	n/a	1	0.98	1	1	1.1	1.2	1	1	1.2	1	1.2	1.2	1	1.9	0.85	0.71	2.2	1

(a) Throughput [×] (green = increases)

		Flink						Storm						Spark						Kafka S.						RisingW.					
		DS2			Dhalion			DS2			Dhalion			DS2			Dhalion			DS2			Dhalion			DS2			Dhalion		
		L	M	H	L	M	H	L	M	H	L	M	H	L	M	H	L	M	H	L	M	H	L	M	H	L	M	H	L	M	H
CPU	Sentiment Analysis (SA)	0.84	0.2	0.53	2.8	2.3	3	3.4	1.3	1.9	4.2	1.1	2.2	0.83	2.4	1.7	0.62	2.2	5.7	0.99	0.74	n/a	0.94	1.1	n/a	1	1.5	1.5	1	1.9	1.5
CPU	Traffic Monitoring (TM)	0.074	0.88	0.18	3	0.84	2.6	0.87	0.47	0.53	1.1	1	1.1	0.18	1.3	2.6	0.97	5	3.3	0.95	n/a	n/a	2.6	1.6	2.1	0.81	1	1	0.36	1	1
CPU	Trending Topics (TT)	1.3	367	1.3	1.5	17	1.3	8.2	1.7	1.3	0.69	3.1	2	4.4	0.22	0.001	2.9	0.98	0.2	3.8	5.3	20	0.12	14	23	0.54	0.67	2.8	1	0.67	6.3
CPU	Machine Outlier (MO)	36	34	10	1.4	15	23	6.1	1.4	0.89	6.6	1.4	1.8	0.96	0.94	1.1	1	0.85	0.76	1	0.081	n/a	1.1	1	n/a	1	1.2	0.99	1	9.3	5.1
S+N State	Ad Analytics (AD)	241	262	178	60	112	26	8.6	0.54	n/a	1.3	1.7	n/a	14	1.1	0.18	3.8	0.63	0.84	0.55	n/a	n/a	10	15	9.2	1.2	1.2	1.2	1.2	1.1	1.2
S+N State	Click Analytics (CA)	844	414	720	480	6.5	60	1.5	1.4	n/a	2.2	1.3	n/a	1.3	0.16	0.87	1.6	1.1	0.99	7.7	n/a	n/a	14	2.6	5.6	1	n/a	0.67	1.5	n/a	1.5
S+N State	Linear Road (LR)	1.1	1.1	0.7	1.1	1.1	1.2	4.7	1.1	6	4.2	0.73	4.3	0.9	0.75	0.79	0.88	0.72	0.61	n/a	n/a	n/a	n/a	0.38	n/a	1.2	1	1.1	1.2	1.1	1.1
State+UDF	Log Analyzer (LA)	644	101	703	50	93	118	4.2	0.97	1.3	3.8	1.3	1.5	3.1	0.9	0.76	11	1.4	1.2	0.28	9.7	n/a	5.2	8.1	n/a	n/a	n/a	n/a	n/a	n/a	n/a
State+UDF	Bargain Index (BI)	8.7	9.7	31	9.7	7.9	4.8	1.8	1.2	n/a	2.5	1.2	n/a	1.7	5.1	0.88	1.9	3.8	0.93	2	1.1	6.5	2	1.1	6.3	2.7	0.99	0.7	2.2	1.4	0.95
State+UDF	Smart Grid (SG)	1	7.3	2.4	20	1.7	17	4.3	1.7	1.9	5.5	1.6	1.5	25	4.5	0.12	33	1.1	0.48	1.6	1.3	n/a	3.9	n/a	n/a	0.8	0.69	0.026	0.81	0.69	0.026
State+UDF	Cloud Monitoring (GCM)	11	0.68	0.58	1.1	3.7	1.2	n/a	n/a	n/a	n/a	n/a	n/a	3.9	1.6	0.14	3.1	1.3	0.52	0.35	56	n/a	1.7	4.5	11	1.2	1.2	1.2	1.2	1.2	1.2
State+UDF	Spike Detection (SD)	9.5	166	93	1.2	28	12	n/a	n/a	n/a	n/a	n/a	n/a	1.5	0.21	1	1.9	1.3	0.51	46	17	15	49	14	12	1.2	1	0.96	1.4	0.98	1.5
Stateless	TPC-H (TPCH)	1.7	1.7	1.1	1.6	1.6	1.8	2.9	1.6	1.3	4.7	1.8	1.7	1	0.75	0.94	1	0.75	0.97	1.2	1.2	0.51	0.91	1	4.2	1.2	1.2	1.2	1.2	1.2	1.2
Stateless	Word Count (WC)	11	4.2	2.1	23	1.2	1.2	3.3	7.7	n/a	1.2	1.3	n/a	0.76	0.74	3.1	0.72	1.2	1.5	0.98	1	4.4	0.95	1.3	4.4	1.8	2.2	2.2	1.8	2.2	2.2

(b) E2E latency [×] (green = decreases)

		Flink						Storm						Spark						Kafka S.						RisingW.					
		DS2			Dhalion			DS2			Dhalion			DS2			Dhalion			DS2			Dhalion			DS2			Dhalion		
		L	M	H	L	M	H	L	M	H	L	M	H	L	M	H	L	M	H	L	M	H	L	M	H	L	M	H	L	M	H
CPU	Sentiment Analysis (SA)	+37	+34	+45	-1	-1	-1	+20	+13	+14	+19	+8	+8	+0	-0	-1	+0	-0	-1	-0	+2	-30	-0	-1	-31	+1	+2	+2	+1	+2	+11
CPU	Traffic Monitoring (TM)	+20	+38	+52	+1	+1	+2	+29	+42	+49	+27	+18	+16	+0	-0	-1	-0	+0	-1	+18	n/a	n/a	+3	-0	+0	+2	+1	+1	+2	+1	+1
CPU	Trending Topics (TT)	+11	+13	+37	+9	+7	+15	-10	+1	+12	-10	+2	-4	+0	+0	-1	+0	+1	-1	+14	+6	+19	+5	+8	+3	+2	-1	+1	+1	+1	+1
CPU	Machine Outlier (MO)	-1	+5	+7	-2	-4	-4	+2	+12	+27	+3	+10	+9	-0	+0	-1	-0	+0	-1	-0	+14	-28	-0	+0	-28	+2	+2	+3	+1	+2	+0
S+N State	Ad Analytics (AD)	-3	-10	-10	-0	-13	-13	+37	+13	+12	+19	+4	+12	-0	+1	-1	-0	+0	-1	+24	n/a	n/a	+16	+8	+3	-1	-0	+1	-1	+0	-0
S+N State	Click Analytics (CA)	-0	-4	-5	-1	-1	-4	+12	+15	+19	+14	+15	+19	+1	-0	-1	+0	+0	-1	+29	n/a	n/a	+25	+18	-17	+1	+2	+1	+1	+2	+1
S+N State	Linear Road (LR)	+1	+1	+3	-0	+0	-1	+29	+32	+27	+26	+20	+16	+0	+0	-1	+0	-0	-1	-1	+3	n/a	+1	+0	-25	-1	+0	-0	-1	-0	+1
State+UDF	Log Analyzer (LA)	+3	+3	-5	+0	-6	-8	-4	-2	+7	-6	-8	-9	+0	+0	-1	+0	+0	-1	+32	+20	-25	+14	+12	-25	n/a	n/a	n/a	n/a	n/a	n/a
State+UDF	Bargain Index (BI)	-2	-3	-4	-3	-5	-4	+4	+28	+20	+3	+20	+20	+0	+0	-1	+0	-0	-1	+5	+3	+3	+1	+1	-3	+0	+0	-0	+1	-1	-0
State+UDF	Smart Grid (SG)	-1	+3	+9	-1	-0	-6	-2	+11	+2	-2	+4	+5	+0	-0	-1	+0	+0	-1	+6	+33	-20	+2	-14	-20	+2	+1	+1	+2	+1	+1
State+UDF	Cloud Monitoring (GCM)	+3	+6	+6	+3	+0	-1	+44	+41	+39	+19	+17	+39	-0	+0	-1	-0	+0	-1	+16	+35	n/a	+14	+3	-6	-1	+0	-0	-0	+1	+0
State+UDF	Spike Detection (SD)	+4	-5	-3	+1	-4	-2	+19	-10	+9	+19	-9	+10	+0	-0	-1	+0	-0	-1	+33	+43	+38	+8	+5	-2	+2	+1	+1	+2	+1	+1
Stateless	TPC-H (TPCH)	+0	+4	+7	+0	+2	+2	-1	+14	+14	-2	+7	+4	+0	+0	-1	+0	-0	-1	-0	+2	-0	-0	+0	-2	-1	-0	+0	-1	+0	+1
Stateless	Word Count (WC)	+1	+1	+2	+1	+0	+1	+28	+34	+36	+26	+24	+36	+0	+0	-1	+0	-0	-1	-1	-0	-0	+0	-1	-2	+1	+2	-1	+2	+2	-0

(c) CPU utilization [pp] (%-points)